Uncertainty-Driven Cyclic Background Purification for Hyperspectral Anomaly Detection

Jicai Li, Yuxin Feng, *Member, IEEE*, Jinqiang Yao, Zhitao Gan, Tao Huang, Zhenyu Wang, Fangfang Wu, Weisheng Dong, *Member, IEEE*, and Guangming Shi, *Fellow, IEEE*

***Abstract*—Hyperspectral anomaly detection (HAD) aims to identify spectrally distinct and spatially sparse targets from complex backgrounds without prior anomaly labels. In recent years, reconstruction-based deep learning (DL) methods have achieved detection by modeling the background distribution. However, due to the influence of imaging conditions, detection networks inevitably reconstruct anomalies alongside the background. Some recent methods indirectly mitigate anomaly overfitting by applying a series of delicate masking strategies, but they usually rely on predefined occlusion rules that are not fully adaptive to diverse anomaly distributions. To address this issue, we propose an uncertainty-driven cyclic framework for hyperspectral anomaly detection (UC-HAD), where pixel-wise reconstruction uncertainty is introduced as a data-adaptive cue to guide background modeling. Within UC-HAD, uncertainty estimation and background reconstruction operate cyclically. Specifically, UC-HAD estimates pixel-wise reconstruction uncertainty by approximating the posterior distribution over model parameters, thereby characterizing reconstruction reliability across spatial locations. Then, an uncertainty-guided adaptive filling strategy (UAFS) is designed. Pixels with low uncertainty are used to adaptively repair high-uncertainty regions, effectively reducing the anomaly interference in subsequent background modeling. As the loop proceeds, the reconstructed background becomes progressively cleaner, and the separability between anomalies and backgrounds is correspondingly enhanced. Experiments on six public HAD datasets demonstrate that UC-HAD consistently outperforms competitive methods, validating the effectiveness of uncertainty-guided cyclic background purification for HAD. The code is available at https://github.com/tobeaTrueman/UC-HAD**

***Index Terms*—Hyperspectral anomaly detection (HAD), hyperspectral image (HSI), image reconstruction, pixel-wise uncertainty.**

Received 29 June 2026; revised 19 August 2026; accepted 24 August 2026. Date of publication 2 September 2026; date of current version 10 September 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62606389, Grant 62472336, Grant 62302369, Grant U2570205, and Grant 62406236; in part by China Postdoctoral Science Foundation under Grant 2024M762551 and Grant 2026M791677; and in part by the Funds of the Natural Science Foundation of Hangzhou under Grant 2025SZRJJ190. *(Corresponding authors: Tao Huang; Yuxin Feng.)*

Jicai Li, Weisheng Dong, and Guangming Shi are with the School of Artificial Intelligence, Xidian University, Xi'an 710071, China.

Yuxin Feng, Zhitao Gan, Tao Huang, and Zhenyu Wang are with Hangzhou Institute of Technology, Xidian University, Hangzhou 311231, China (e-mail: fyxxq0126@gmail.com; httz1001@163.com).

Jinqiang Yao is with CICO Research Global, Hangzhou 310020, China.

Fangfang Wu is with the School of Computer Science and Technology, Xidian University, Xi'an 710071, China.

This article has supplementary downloadable material available at https://doi.org/10.1109/TGRS.2026.3729637, provided by the authors.

Digital Object Identifier 10.1109/TGRS.2026.3729637

I. Introduction

HYPERSPECTRAL image (HSI) interpretation plays a crucial role in remote sensing [1], [2], as it exploits rich spectral information to characterize subtle material differences. An important task in this domain is hyperspectral anomaly detection (HAD), which aims at identifying spectrally deviant pixels from complex backgrounds without prior knowledge of anomaly categories. Due to its ability to discover rare and unknown targets, HAD has been widely applied in military surveillance, agriculture monitoring, and transportation inspection. However, its performance is fundamentally constrained by two inherent bottlenecks. First, anomalies exhibit extreme spatial scarcity [3], typically occupying only 0.0019%–0.48% of the total image area. Second, the lack of pixel-level labels forces HAD to operate in an unsupervised manner. These constraints severely hinder the development and practical deployment of HAD methods.

Over the past decades, numerous algorithms have been proposed for HAD [1], [2]. Among them, statistics-based models [4], [5], [6] are the most representative traditional approaches. These methods usually assume that background pixels follow a specific statistical distribution and regard pixels deviating from this distribution as anomalies. However, such assumptions are often difficult to satisfy in real hyperspectral scenes with complex and heterogeneous backgrounds, which limits detection performance. Recently, deep learning (DL) has attracted increasing attention in hyperspectral image analysis [7], [8] because of its strong representation capability. In particular, reconstruction-based deep models [9], [10], [11] have shown promising performance for HAD. These methods are built on the assumption that dominant background components can be effectively modeled by deep networks, so that a clean background can be reconstructed from the input hyperspectral image. Nevertheless, anomalous pixels tend to be partially reconstructed as the background by the network, which causes diminished separability between anomalies and normal background and thus degrades detection performance.

To alleviate this problem, existing reconstruction-based HAD methods have explored different anomaly-suppression mechanisms. The autoencoder (AE) [11], [12], GAN, and diffusion-based methods [9], [13] enhance background reconstruction or background suppression through robust reconstruction constraints, adversarial learning, or generative denoising. More recently, mask-based [14], [15] and blind-

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:15:16 UTC from IEEE Xplore. Restrictions apply.

spot [16], [17] strategies have achieved promising results by preventing the network from directly copying anomalous regions and forcing it to recover masked or blind-spot areas from surrounding spectral–spatial context.

Despite these advances, existing strategies still rely heavily on externally designed objectives, constraints, or reconstruction rules. For AE, GAN, and diffusion-based methods, their performance is often affected by carefully designed reconstruction objectives, adversarial constraints, or generative assumptions. More importantly, masking and blind-spot-based strategies introduce additional masking configurations, such as mask regions, scales, ratios, or patterns. Due to the sparse, irregular, and scene-dependent nature of anomalous targets, these configurations are difficult to determine and often require careful parameter tuning. Although random masking strategies have been explored to reduce handcrafted designs [14], [18], they remain externally imposed and data-agnostic, which may limit their adaptability to heterogeneous anomaly distributions [19], [20]. In contrast to these externally designed anomaly-suppression mechanisms, the model's inherent reconstruction reliability, which can be characterized by pixel-wise uncertainty, has been largely overlooked.

Motivated by the above analysis, we reconsider the key challenge in the reconstruction-based HAD paradigm. Existing masking-based methods suppress anomaly leakage by imposing predefined occlusion rules, yet they do not explicitly characterize whether each spatial location is reliably reconstructed by the current model. In hyperspectral images, anomalous pixels usually exhibit spectral responses that deviate from surrounding backgrounds, whereas dominant background regions are generally more homogeneous and easier to model [21]. As a result, background regions tend to be reconstructed more consistently, while anomalous or difficult-to-explain regions are more likely to yield unstable reconstruction behaviors. This observation suggests that the reliability of background reconstruction can be evaluated in a data-adaptive manner. To this end, reconstruction uncertainty provides a natural cue for measuring how confidently each spatial location is explained by the current model. Compared with mask-based methods that rely on predefined occlusion patterns, uncertainty estimation offers a more direct and adaptive way to identify unreliable regions during background modeling. The conceptual difference between these two detection paradigms is illustrated in Fig. 1.

Building on the above insight, we propose an uncertainty-driven cyclic framework for hyperspectral anomaly detection (UC-HAD). The core idea is to treat pixel-wise reconstruction uncertainty as an intrinsic anomaly-aware reliability cue and use it to guide data-adaptive and iterative background purification. Specifically, we estimate pixel-wise reconstruction uncertainty by performing multiple stochastic forward passes, from the perspective of approximating the posterior distribution over model parameters. High-uncertainty regions are regarded as reconstruction-unreliable regions that may contain anomalous pixels, while low-uncertainty regions are regarded as more reliable background components. Based on this uncertainty estimation, a candidate pool of reliable background pixels is constructed from low-uncertainty regions. The elements in this candidate pool are then used to adaptively repair high-uncertainty regions, thereby suppressing anomalous interference in subsequent background modeling. The repaired hyperspectral image is then fed into a background reconstructor (BR) to generate a cleaner background representation, and the reconstructed result is further used to update pixel-wise uncertainty. Through this cyclic interaction between uncertainty estimation and background refinement, the reconstructed background becomes progressively cleaner and more discriminative for anomaly detection. Finally, the anomaly map is obtained from the residual between the input hyperspectral image and the reconstructed background.

Our proposed UC-HAD framework is evaluated on six publicly available HAD datasets. Experimental results show that UC-HAD consistently outperforms competitive methods in terms of AUC and delivers robust detection performance across diverse hyperspectral remote sensing scenes. The main contributions of this article can be summarized as follows.

1) We introduce pixel-wise reconstruction reliability into reconstruction-based HAD, where reconstruction uncertainty is used as an anomaly-aware cue beyond conventional reconstruction residuals.
2) We propose an uncertainty-guided adaptive filling strategy (UAFS) for spectral–spatial background purification, which repairs unreliable regions using reliable background evidence.
3) We develop an uncertainty-driven cyclic framework that couples uncertainty estimation, adaptive filling, and background reconstruction to progressively obtain a cleaner pseudo-background.

The rest of this article is organized as follows. Section II reviews related work on the HAD task. Section III describes the proposed UC-HAD framework. Section IV reports the experimental results and analysis. Section V concludes the article with a summary of the main findings, limitations of the proposed method, and possible future research directions.

II. Related Work

A. Traditional HAD Methods

Traditional HAD methods mainly include statistics-based models [5], [22] and representation-based models [23], [24], [25], [26]. Statistical-based methods model the background of HSIs and quantify anomalies by measuring deviations from the background distribution. The most representative method is the Reed–Xiaoli (RX) detector [4], which models the background as a multivariate Gaussian distribution and adopts the Mahalanobis distance as the anomaly criterion. Based on RX, numerous extensions [27], [28], [29], [30], [31] have been developed to address complex scenarios, such as kernel RX-AD [32] for nonlinear distributions, spatially adaptive weighted RX-AD [33] and RX-AD variants with linear filtering for enhanced background suppression. Although statistically well grounded, these methods are highly sensitive to the accuracy of background distribution assumptions.

Representation-based methods exploit intrinsic properties of HSIs to construct anomaly detection models. According to the employed priors, they can be categorized into sparse representation [23], [24], [34], [35], collaborative representation [25], and low-rank decomposition. Li et al. [36] proposed a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:15:16 UTC from IEEE Xplore. Restrictions apply.

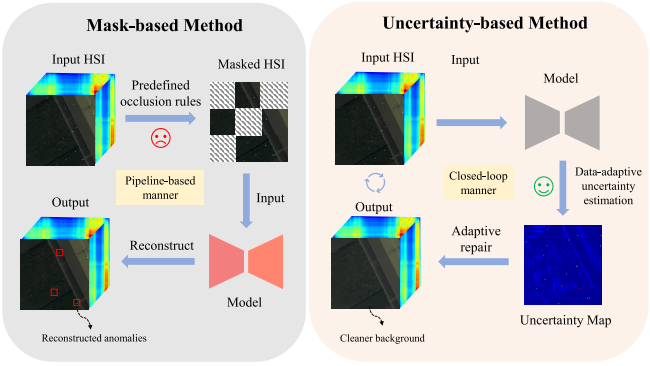


Fig. 1. Comparison between pipeline-based mask reconstruction and uncertainty-driven cyclic reconstruction for HAD. Mask-based reconstruction recovers predefined or randomly masked regions through a fixed reconstruction pipeline, whose effectiveness depends on the manually designed masking strategy. In contrast, UC-HAD estimates pixel-wise reconstruction uncertainty to identify unreliable regions, uses reliable background evidence for adaptive repair, and progressively purifies the background through cyclic refinement.

sparse representation-based detector that identifies anomalies through sparse reconstruction. The classical collaborative representation detector (CRD) [37] localizes anomalies by the reconstruction error, while the low-rank and sparse matrix decomposition (LSDM) [38] model addresses anomaly detection by decomposing HSIs into low-rank background and sparse anomaly components. Despite relaxing strict statistical assumptions, representation-based methods still rely heavily on handcrafted priors, which limits their adaptability in complex scenes.

B. Deep-Learning-Based Methods

In recent years, DL has become a prominent paradigm for HSI analysis [7], [8], owing to its strong ability to extract high-level representations, and has been widely adopted in HAD. In unsupervised HAD, a prevalent paradigm is to formulate anomaly detection as a background modeling problem, where background-dominant patterns are learned from unlabeled HSIs and anomalies are identified according to their deviations from the estimated background. Wang et al. [11] introduced a representative autonomous HAD framework, termed Auto-AD, where an AE is employed to reconstruct the background, while anomalies are identified through the resulting reconstruction residuals. In addition to AE-based models, GAN and diffusion-based models [9], [13], [39], [40] have also been explored for HAD. These methods enhance background modeling and anomaly discrimination by learning complex data distributions, but they may still suffer from unstable training and high computational complexity. Recently, sequence-modeling-based methods [10], [41], [42] have also been introduced into HAD to enhance the modeling of long-range spectral–spatial dependencies. D3Former [43] adopts a dual-decoder dual-Transformer reconstruction network to improve background feature purification and strengthen anomaly–background separability. In addition, Mamba-based methods [44], [45] have also attracted increasing attention because their state-space modeling mechanism enables efficient long-range dependency modeling in high-dimensional HSIs with long spectral sequences.

In parallel with these advances, hyperspectral large models and foundation models [46], [47], [48] have provided a new paradigm for HSI interpretation. HyperSIGMA [49] introduces a large-scale vision foundation model specifically designed for hyperspectral imagery, where large-scale pretraining is employed to learn transferable spectral–spatial representations across diverse downstream tasks. Such hyperspectral foundation models show strong potential for HAD and HTD [50], [51], [52] by providing generalizable feature representations and improving adaptation to different scenes [19], [20]. However, most existing large-model-based studies mainly focus on enhancing general representation learning and transferability, while their fine-grained perception of anomalous regions in HAD remains insufficiently explored.

C. Uncertainty Theory

Uncertainty describes the credibility of a model's predictions when the available information is incomplete or ambiguous [53], [54]. In the broader fields of computer vision and machine learning, uncertainty quantification has been established as a cornerstone of credible artificial intelligence. Kendall and Gal [55] formally proposed a unified framework for capturing accidental and cognitive uncertainties in Bayesian DL and demonstrated its performance improvement in semantic segmentation and regression tasks. To address the computational complexity of Bayesian methods, Lakshminarayanan et al. [56] introduced "deep ensembles," which estimate prediction uncertainty by training multiple independently initialized neural networks. Gal and Ghahramani [57] provided a theoretical basis for using Monte Carlo (MC) dropout as a Bayesian approximation, enabling standard neural networks to effectively model uncertainty during inference. Ebel et al. [58] proposed a probabilistic framework for cloud removal in optical satellite time series. They demonstrated that by explicitly modeling uncertainty, the network can distinguish between clear observation data and damaged data, thus significantly enhancing the robustness of reconstruction to atmospheric disturbances. However, despite significant progress in uncertainty modeling in DL and computer vision, most HAD methods still follow a purely deterministic paradigm. Existing detectors largely rely on a single reconstruction error to identify anomalies, ignoring the reliability of such predictions.

III. Method

In this section, we first define the HAD task in Section III-A. Then, Section III-B provides a theoretical analysis of reconstruction uncertainty for background reliability modeling, revealing that reconstruction uncertainty can characterize the spatial reliability of background reconstruction. Based on this analysis, Section III-C introduces a UAFS for unreliable regions, which aims to progressively suppress anomalous interference in background modeling. In Section III-D, we present the complete UC-HAD, which consists of an uncertainty estimator (UE) and a BR operating cyclically. Finally, the optimization objective of the UC-HAD is also formulated in Section III-E.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:15:16 UTC from IEEE Xplore. Restrictions apply.

A. Problem Formulation

HAD aims to identify pixel-level anomalous regions whose spectral characteristics deviate from the background. In reconstruction-based HAD, background reconstruction is commonly adopted as an auxiliary task, and anomaly detection is performed by measuring the residual between the original hyperspectral image and the reconstructed background. Our method follows this paradigm. Formally, the HAD process can be written as follows:

$$\begin{aligned} X_B &= F(\mathbf{X}) \\ X_A &= \mathbf{X} - X_B \end{aligned} \tag{1}$$

where $\mathbf{X} \in \mathbb{R}^{H\times W\times B}$ denotes the input hyperspectral image, where H, W, and B represent the spatial height, spatial width, and number of spectral bands, respectively. $F(\cdot)$ denotes the background reconstruction network, $\mathbf{X}_B$ is the reconstructed background, and $\mathbf{X}_A$ denotes the anomaly residual.

B. Reconstruction Uncertainty for Background Reliability Modeling

A fundamental, yet often overlooked aspect in reconstruction-based HAD is that the reliability of background reconstruction is spatially heterogeneous. Instead of directly characterizing anomalies, we revisit HAD from the perspective of reconstruction reliability, because reliable background modeling provides a more stable basis for subsequent background purification and anomaly detection. In reconstruction-based methods, the network is usually trained to reconstruct the input hyperspectral image by fitting its dominant components. Since background pixels occupy the major part of a scene and generally exhibit more homogeneous spectral patterns, the learned model tends to form a stable representation of the background distribution. In contrast, anomalous or difficult-to-explain regions deviate from this dominant distribution and are, therefore, less likely to be reconstructed consistently. As a result, different spatial locations are not equally explainable by the learned model, and their reconstruction reliability varies accordingly.

To characterize this reliability, it is insufficient to rely only on deterministic reconstruction outputs, because a single prediction cannot reveal whether a region is stably explained by the current model. From a probabilistic perspective, reconstruction reliability can be associated with the stability of predictions under plausible model hypotheses. Given an input pixel x, the predictive distribution of its reconstruction $\hat{x}$ can be formulated as

$$p(\hat{x} \mid x, X) \approx \int p(\hat{x} \mid x, \theta)\, q(\theta)\, d\theta \tag{2}$$

where $\hat{x}$ denotes the reconstructed output of pixel x, θ represents the model parameters, and $q(\theta)$ denotes the posterior distribution of the model parameters. Under this formulation, regions that can be stably explained by the learned background model tend to yield consistent predictions, whereas anomalous or difficult regions are more likely to produce unstable predictions. Thus, reconstruction uncertainty serves as a pixel-wise reliability indicator rather than a simple postprocessing score.

However, directly obtaining the true posterior distribution $q(\theta)$ is generally intractable. Therefore, we adopt an approximate inference method inspired by MC Dropout [57] to make the problem tractable in a computationally efficient manner. Specifically, the model parameters are interpreted as stochastic samples from an implicit posterior

$$\theta^{(k)} \sim q(\theta) \tag{3}$$

where k denotes the stochastic sampling index. By performing multiple stochastic forward passes, the model produces a set of plausible reconstructions under different parameter instantiations

$$\hat{x}^{(k)} = F_{\theta^{(k)}}(x),\ k = 1, \ldots, T \tag{4}$$

where T denotes the total number of stochastic forward passes, and $F_{\theta^{(k)}}(\cdot)$ denotes the reconstruction network parameterized by $\theta^{(k)}$. Based on this formulation, we define the reconstruction uncertainty of pixel x as the predictive variance

$$\mathcal{U}(x) = \mathrm{Var}_{\theta\sim q(\theta)}\left[F_\theta(x)\right]. \tag{5}$$

The variance quantifies the sensitivity of the reconstruction to parameter perturbations. Unlike deterministic reconstruction errors, this measure reflects the consistency of model predictions in the hypothesis space, thereby serving as a principled indicator of reconstruction reliability. Here, x represents the complete spectral vector at a given spatial location rather than an independent scalar response.

It is important to emphasize that, in our method, uncertainty is not directly treated as the anomaly score. This is because high uncertainty does not necessarily indicate anomaly, especially in regions with complex background distributions. Instead, it serves as an intermediate reliability cue for identifying unreliable regions and guiding subsequent background purification in a data-adaptive manner. Compared with mask-based strategies that impose external perturbations in the input space, the proposed uncertainty-aware mechanism provides a more intrinsic and adaptive way to characterize reconstruction reliability. Such uncertainty characterization will be further exploited in Section III-C to guide background purification.

C. Uncertainty-Guided Adaptive Filling Strategy

The overall framework of UC-HAD is illustrated in Fig. 2. Instead of suppressing anomalous interference through predefined occlusion rules as in masking-based methods, Wu et al. [18], and Huang et al. [59], we propose an UAFS to purify background reconstruction in a data-adaptive manner, as shown in Fig. 3. The key idea is to use reconstruction uncertainty as an intermediate reliability cue to locate unreliable regions. These regions are then adaptively repaired with reliable background evidence, thereby reducing anomaly contamination in subsequent background modeling.

Specifically, we first utilize the uncertainty map $\mathcal{U}$ to generate an unreliable-region indicator I

$$I(x,y) = \begin{cases} 1, & \text{pixels with top 5\% of } \mathcal{U}(x,y) \\ 0, & \text{otherwise} \end{cases} \tag{6}$$

where pixels with $I(x,y) = 1$ denote the top 5% high-uncertainty pixels in the uncertainty map. These pixels

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:15:16 UTC from IEEE Xplore. Restrictions apply.

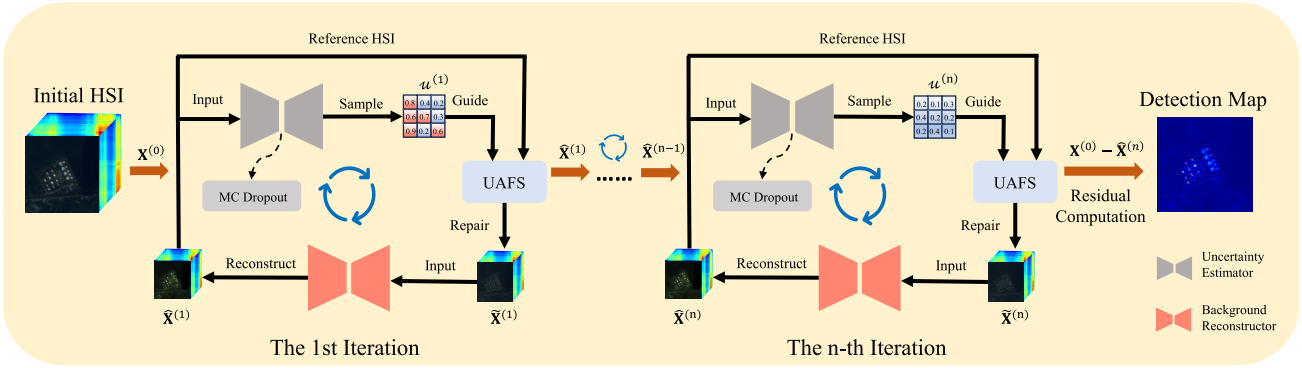


Fig. 2. Overall framework of UC-HAD. The framework consists of an UE, an UAFS module, and a BR, which interact cyclically to progressively purify background reconstruction.

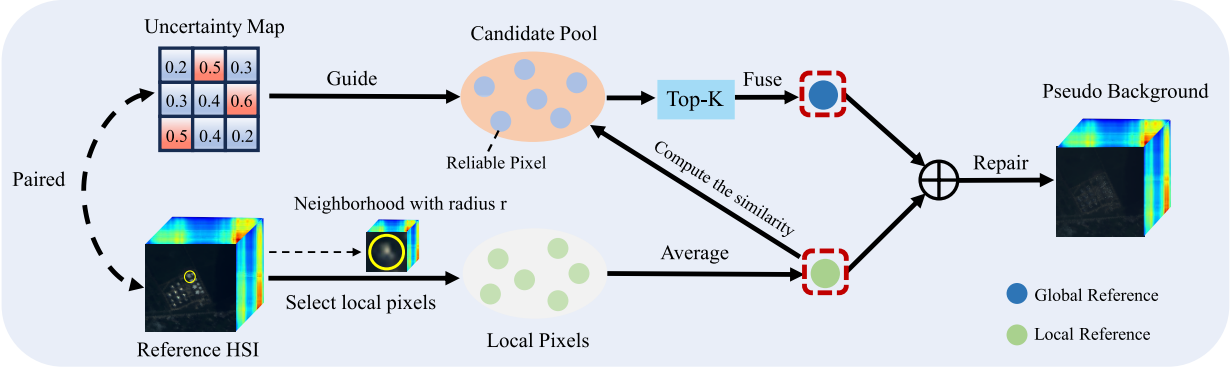


Fig. 3. Workflow of the proposed UAFS. High-uncertainty regions are repaired for cleaner background reconstruction.

correspond to regions that are not reliably explained by the current reconstruction model and may include both anomalous pixels and difficult-to-model background pixels. Therefore, the indicator I is used only to locate regions requiring adaptive repair.

We then select low uncertainty pixels from $\mathcal{U}(x, y)$ to construct a candidate pool $\mathcal{P}$ of reliable background samples

$$\mathcal{P} = [\mathbf{s}_1, \mathbf{s}_2, \mathbf{s}_3, \ldots, \mathbf{s}_n] \tag{7}$$

where $\mathbf{s}_i$ denotes the ith element in the candidate pool $\mathcal{P}$, and n is the number of candidate samples. Since these pixels are reconstructed with higher confidence, they provide reliable background evidence for adaptive repair.

For each high uncertainty pixel, we compute the local mean $\boldsymbol{\mu}_{\text{local}}$ of low uncertainty pixels within a neighborhood of radius r as the local background reference. We then compute the similarity weight between the local reference and each candidate $\mathbf{s}_i \in \mathcal{P}$ as

$$w_i = \exp\left(-\frac{\left\|\mathbf{s}_i - \boldsymbol{\mu}_{local}\right\|^2}{2\sigma_{spec}^2}\right) \tag{8}$$

where σ_{spec} is a hyperparameter controlling the scale of the similarity.

The top-K candidates in $\mathcal{P}$ with the highest weights are selected to obtain a global background reference $\mathbf{b}_{\text{global}}$

$$\mathbf{b}_{\text{global}} = \frac{\sum_{i\in\mathcal{K}} w_i \mathbf{s}_i}{\sum_{i\in\mathcal{K}} w_i}. \tag{9}$$

Finally, the repaired pixel $\tilde{\mathbf{x}}(x, y)$ is obtained by linearly fusing the global background reference and the local background reference

$$\tilde{\mathbf{x}}(x, y) = \alpha \mathbf{b}_{global} + (1 - \alpha)\boldsymbol{\mu}_{\text{local}} \tag{10}$$

where α is a fusion factor that controls the relative contributions of global background evidence and local spatial context.

Through this process, high-uncertainty regions are progressively repaired with reliable background evidence, thereby reducing the tendency of the subsequent network to reconstruct anomalous patterns and yielding more faithful and consistent background modeling. Unlike generic masking or filling operations, UAFS derives the reliability map from reconstruction uncertainty and exploits both spectral similarity and spatial context for background purification. This strategy differs fundamentally from hand-crafted masking-based methods, which heavily rely on predefined occlusion patterns and do not explicitly model reconstruction reliability in a data-adaptive manner.

D. Uncertainty-Driven Cyclic Framework

As illustrated in Fig. 2, the proposed UC-HAD consists of an UE, an uncertainty-guided adaptive filling module, and a BR. These components do not operate independently in a single pass. Instead, they interact cyclically, so that unreliable regions can be progressively repaired and background reconstruction can be iteratively purified.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:15:16 UTC from IEEE Xplore. Restrictions apply.

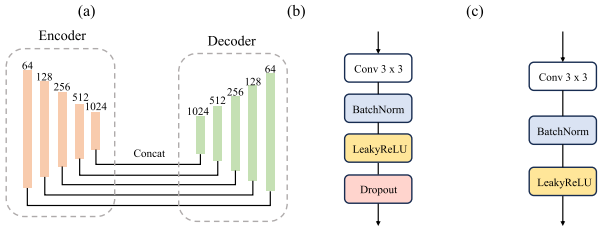


Fig. 4. Detailed network architecture of the proposed UC-HAD. (a) Architecture of the U-Net. (b) Convolutional block structure of the UE. (c) Convolutional block structure of the BR.

Algorithm 1 Cyclic Refinement Mechanism

Input original HSI $\mathbf{X}^{(0)} \in \mathbb{R}^{H\times W\times B}$, convergence tolerance η, small constant ϵ, stochastic pass number T
Initialize $t = 0$
repeat
  Estimate uncertainty map $\mathcal{U}^{(t)}$ using UE on $\mathbf{X}^{(t)}$ with T stochastic forward passes
  Generate pseudo-background $\tilde{\mathbf{X}}^{(t)} \leftarrow \mathcal{M}(\mathbf{X}^{(t)}, \mathcal{U}^{(t)})$
  Reconstruct candidate background $\hat{\mathbf{X}}^{(t)} \leftarrow f_{\theta_B}^{(t)}(\tilde{\mathbf{X}}^{(t)})$
  Evaluate the candidate uncertainty reduction $\Delta_{\text{cand}}^{(t+1)}$
  Accept $\mathbf{X}^{(t+1)} \leftarrow \hat{\mathbf{X}}^{(t)}$ if $\Delta_{\text{cand}}^{(t+1)} \geq \eta$;
  $t \leftarrow t + 1$ if the candidate update is accepted
until the candidate update is rejected
Output final accepted background $\mathbf{X}^{(t)}$

1) Network Architecture: In UC-HAD, UE is responsible for characterizing pixel-wise reconstruction reliability through stochastic predictions, while BR aims to generate a cleaner and more stable background representation for anomaly detection. Specifically, both the UE and BR are implemented based on a U-Net architecture with an encoder–decoder structure and skip connections. The encoder consists of four downsampling stages, and the channel numbers are set to 64, 128, 256, 512, and 1024 from shallow to deep layers. The decoder follows a symmetric structure to progressively recover the spatial resolution and integrates multiscale encoder features through skip connections. Each convolutional block contains a 3×3 convolution, batch normalization, and LeakyReLU activation. The main difference between the UE and BR lies in the use of Dropout. In the UE, Dropout layers with a dropout rate of 0.2 are inserted into the convolutional blocks, so that multiple stochastic forward passes can be performed on the same input during inference. In contrast, BR adopts the same U-Net backbone but removes the Dropout layers to provide deterministic reconstruction and more stable fitting of dominant background components. The detailed structures of the UE and BR are illustrated in Fig. 4.

2) Cyclic Refinement Mechanism: A key motivation of the proposed cyclic refinement mechanism is that a single round of uncertainty-guided filling is generally insufficient to remove anomalous interference from background modeling. This is because the initial uncertainty map is estimated from the original hyperspectral image, which still contains anomalous components and complex background interference. As a result, the initial pseudo-background remains imperfect, and the corresponding reconstruction cannot fully reflect the clean background distribution. To address this issue, UC-HAD alternates uncertainty estimation, uncertainty-guided adaptive filling, and background reconstruction over multiple iterations, so that reconstruction reliability and background purity can be progressively improved. This cyclic process forms an uncertainty-driven feedback loop rather than a one-pass cascade, progressively shifting reconstruction from the contaminated HSI toward a cleaner pseudo-background. The complete refinement procedure is summarized in Algorithm 1.

At the tth iteration, the UE first performs pixel-wise reconstruction on the current input $\mathbf{X}^{(t)}$ and produces an uncertainty map $\mathcal{U}^{(t)}$ through T stochastic forward passes

$$\mathcal{U}^{(t)}(x,y) = \frac{1}{BT}\sum_{b=1}^{B}\sum_{k=1}^{T}\left(\widehat{X}_{k,b}^{(t)}(x,y) - \overline{X}_{b}^{(t)}(x,y)\right)^2 \tag{11}$$

where $\overline{X}_{b}^{(t)}(x,y) = 1/T\sum_{k=1}^{T}\widehat{X}_{k,b}^{(t)}(x,y)$ denotes the predictive mean of the bth spectral band at spatial location (x,y), $\widehat{X}_{k,b}^{(t)}(x,y)$ denotes its reconstructed value under the kth stochastic forward pass, and B and T denote the number of spectral bands and stochastic forward passes, respectively.

Based on $\mathcal{U}^{(t)}$, the UAFS is then applied to obtain a pseudo-background image

$$\tilde{\mathbf{X}}^{(t)} = \mathcal{M}\left(\mathbf{X}^{(t)}, \mathcal{U}^{(t)}\right) \tag{12}$$

where $\mathcal{M}(\cdot)$ denotes the uncertainty-guided filling operation. Compared with the original input, $\tilde{\mathbf{X}}^{(t)}$ suppresses unreliable regions while preserving reliable background components.

The resulting pseudo-background is subsequently fed into BR to obtain a cleaner reconstruction

$$\hat{\mathbf{X}}^{(t)} = f_{\theta_B}^{(t)}\left(\tilde{\mathbf{X}}^{(t)}\right) \tag{13}$$

where $f_{\theta_B}^{(t)}(\cdot)$ denotes the BR at the tth iteration. Since the training input has been progressively purified, the BR is able to learn a more stable and accurate representation of the underlying background distribution. The reconstructed output $\hat{\mathbf{X}}^{(t)}$ is then used as the input of the UE in the $(t+1)$th iteration.

3) Convergence Mechanism: Through the alternating update process, unreliable regions are progressively repaired with background-consistent information, while reliable background regions become increasingly stable across iterations. Therefore, the cyclic refinement process can be regarded as an uncertainty-reduction process. We adopt an adaptive stopping criterion based on the relative reduction of the overall reconstruction uncertainty between two consecutive cycles.

Specifically, after the tth cycle, the overall reconstruction uncertainty score is computed as

$$S^{(t)} = \sum_{i=1}^{H}\sum_{j=1}^{W}\mathcal{U}^{(t)}(i,j) \tag{14}$$

where $\mathcal{U}^{(t)}$ denotes the pixel-wise uncertainty map in the tth cycle, and H and W denote the spatial size of the HSI. Then, the relative uncertainty reduction is defined as

$$\Delta_{\text{cand}}^{(t+1)} = \frac{S^{(t)} - S_{\text{cand}}^{(t+1)}}{S^{(t)} + \epsilon} \tag{15}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:15:16 UTC from IEEE Xplore. Restrictions apply.

where $S_{\text{cand}}^{(t+1)}$ denotes the raw overall uncertainty of the candidate reconstruction. In our implementation, ϵ and η are fixed to 10^{-8} and 0.01, respectively, for all datasets.

The candidate update is accepted only when $\Delta_{\text{cand}}^{(t+1)} \geq \eta$; otherwise, it is rejected, and the current reconstruction is retained as the final background estimate. For each accepted update, we have $S^{(t+1)} < S^{(t)}$. Since $S^{(t)} \geq 0$, the retained uncertainty sequence is monotonically decreasing and bounded below. The detailed mathematical derivation and experimental validation are provided in the Supplementary Material.

E. Training and Testing

In UC-HAD, both the UE and BR are optimized using mse reconstruction losses, but the losses are applied to different input–output pairs according to their respective roles. The UE is used to estimate pixel-wise reconstruction uncertainty through stochastic predictions, whereas the BR is used to learn a cleaner and more stable background representation for anomaly detection. For the UE, the input is the current HSI patch $x_i^{(t)}$, and the output is its reconstructed patch $\hat{x}_{U,i}^{(t)} = f_{\theta_U}(x_i^{(t)})$. The training objective of the UE is formulated as

$$\mathcal{L}_{UE} = \frac{1}{N}\sum_{i=1}^{N}\left\|\hat{x}_{U,i}^{(t)} - x_i^{(t)}\right\|_2^2. \tag{16}$$

After training, the UE is used to estimate pixel-wise reconstruction uncertainty through stochastic forward passes. For the BR, the input is the pseudo-background patch $\tilde{x}_i^{(t)}$ generated by UAFS, and the output is the reconstructed background patch $\hat{x}_{B,i}^{(t)} = f_{\theta_B}(\tilde{x}_i^{(t)})$. The training objective of the BR is defined as

$$\mathcal{L}_{BR} = \frac{1}{N}\sum_{i=1}^{N}\left\|\hat{x}_{B,i}^{(t)} - \tilde{x}_i^{(t)}\right\|_2^2. \tag{17}$$

During testing, the final trained BR is used to reconstruct the background of the input hyperspectral image. The anomaly map is then obtained as the residual between the input image and the reconstructed background

$$\mathbf{X}_A = \left|\mathbf{X} - \hat{\mathbf{X}}\right| \tag{18}$$

where $\hat{\mathbf{X}}$ denotes the reconstructed background generated by the BR, and $\mathbf{X}_A$ represents the final anomaly response.

IV. Experiments and Analysis

A. Experimental Settings

1) Datasets: In the experiments, the proposed UC-HAD was validated on the six public datasets including Salinas [60], Texas Coast [60], Bay Champagne [60], HSI-II [60], AVIRIS-I [60], and HYDICE [11]. Details of each dataset are shown in Table I.

2) Comparison Models and Evaluation Metrics: To verify the effectiveness of our proposed method, we compared UC-HAD with representative traditional, representation-based, and deep-learning-based HAD methods, including the statistics-based LRX [61], the representation-based CRD [37], and several DL-based methods, namely Auto-AD [11], AETNET [14], MSNET [62], TDD [3], CBAR [63], NL2NET [64], SuperAD [65], and the recent Mamba-based method SSHAD [66]. Among them, AETNET adopts a random-mask-based training strategy to enhance anomaly sensitivity.

TABLE I
Introduction of the Six Public Datasets

Dataset	Sensor	Spectral range (nm)	Size	Anomalies
Salinas	AVIRIS	400-2500	120×120×224	Roof
Texas Coast	AVIRIS	400-2500	100×100×207	Buildings
Bay Champagne	AVIRIS	400-2500	100×100×188	Watercraft
HSI-II	ROSIS-03	430-2500	100×100×193	Vehicles
AVIRIS-I	AVIRIS	350-2510	100×100×186	Airplanes
HYDICE	HYDICE	400-2500	80×100×175	Roof, Cars

Following recent HAD evaluation protocols, we adopt the 3-D ROC [67], [68] metric to comprehensively evaluate anomaly detection capability, background suppression performance, and overall detection robustness. Compared to the conventional metric $\text{AUC}_{(D,F)}$, 3-D ROC analysis incorporates the detection threshold τ as an additional dimension, thereby providing a more rigorous assessment of both anomaly detectability and background suppressibility. Specifically, $\text{AUC}_{(D,F)}$ measures the overall discriminative capability, while $\text{AUC}_{(D,\tau)}$ and $\text{AUC}_{(F,\tau)}$ evaluate the accumulated detection response and false alarm tendency under varying thresholds, respectively. In addition, AUC_{ODP} and AUC_{JAD} are used as comprehensive evaluation metrics, AUC_{SNPR} is employed to measure the tradeoff between detection performance and false alarm suppression, and AUC_{ADBS} is adopted to specifically assess the ability to suppress background clutter.

3) Implementation Details: For LRX and CRD, the inner and outer window sizes are set to 5/9 and 11/17, respectively. For TDD, we follow the official implementation and use the pretrained weights [3]. Other compared methods are evaluated using their official implementations with default or recommended parameter settings, without dataset-specific label-based tuning. This is because HAD is generally conducted in an unsupervised setting, where using ground-truth anomaly labels to exhaustively tune each competing method on each test scene would introduce dataset-specific supervision and lead to unfair over-optimization. For the proposed UC-HAD, each input hyperspectral image is cropped into patches of size 32×32 with a stride of 8. During training, the Adam optimizer is adopted with an initial learning rate of 1×10^{-3}. The UE and the BR are trained for 150 and 200 epochs, respectively. The dropout rate is set to 0.2, and the number of MC stochastic forward passes is set to $T = 5$. A fixed random seed is used for stochastic sampling to ensure reproducibility. All experiments are conducted on a workstation equipped with an Intel Xeon Gold 6226R CPU at 2.90 GHz and an NVIDIA RTX 3090 GPU with 24 GB memory. More detailed experimental configurations are provided in the Supplementary Material.

B. Model Comparison

1) Detection Performance Comparison: UC-HAD is compared with multiple HAD models, and the quantitative

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:15:16 UTC from IEEE Xplore. Restrictions apply.

TABLE II
QUANTITATIVE COMPARISON RESULTS ON THE SALINAS AND TEXAS COAST DATASETS

Model	Salinas							Texas Coast						
	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{JAD}	AUC_{ADBS}	AUC_{SNPR}	AUC_{OADP}	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{JAD}	AUC_{ADBS}	AUC_{SNPR}	AUC_{OADP}
LRX	0.9292	0.2028	0.0415	1.1200	0.1567	4.8772	2.0803	0.9965	0.2832	0.0330	1.2751	0.2413	8.4861	2.2434
CRD	0.9677	0.1638	0.0352	1.1417	0.1325	4.5732	2.1110	0.9428	0.3090	0.0447	1.2555	0.2477	6.6722	2.2150
Auto-AD	0.9908	0.1421	0.0323	1.1315	0.1128	4.3720	2.0936	0.9963	0.3814	0.0343	1.3728	0.3426	11.1721	2.3480
AETNET	0.9292	0.1886	0.0311	1.3178	0.1925	18.4084	2.2967	0.9935	0.0985	0.0747	1.0920	0.0854	13.1291	2.0845
TDD	0.5356	0.0628	0.0594	0.5736	0.0115	0.7645	0.5293	0.1824	0.4504	0.3509	0.6303	0.1098	1.2903	1.2845
CBAR	0.9568	0.1887	0.0376	1.2483	0.1511	5.0186	2.2018	0.9863	0.1518	0.0287	1.2085	0.1231	5.2892	2.1937
NL2NET	0.9729	0.1808	0.0257	1.4255	0.1526	27.0410	2.4193	0.9907	0.1312	0.0285	1.2639	0.0133	14.4281	2.2594
SSHAD	0.9874	0.1835	0.0258	1.7826	0.1577	7.1124	2.7421	0.9985	0.2494	0.0178	1.2568	0.2316	14.0112	2.2520
SuperAD	0.9593	**0.2075**	0.0289	1.3628	**0.1734**	7.1799	2.4165	0.9983	0.1312	0.0218	1.2149	0.1094	6.0183	2.2386
UC-HAD	**0.9926**	0.1854	**0.0255**	**2.0638**	0.1602	**30.9633**	**3.0362**	**0.9988**	**0.2749**	**0.0151**	**1.2763**	**0.2612**	**18.3549**	**2.2612**

TABLE III
QUANTITATIVE COMPARISON RESULTS ON THE BAY CHAMPAGNE AND HSI-II DATASETS

Model	Bay Champagne							HSI-II						
	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{JAD}	AUC_{ADBS}	AUC_{SNPR}	AUC_{OADP}	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{JAD}	AUC_{ADBS}	AUC_{SNPR}	AUC_{OADP}
LRX	0.9965	0.2887	0.0309	1.2725	0.2467	8.4816	2.2455	0.9588	0.1099	0.0559	1.0587	0.0477	1.8188	2.0054
CRD	0.9567	0.3089	0.0451	1.2509	0.2530	6.6799	2.2198	0.9425	0.1090	0.0468	1.0405	0.0533	2.1708	1.9919
Auto-AD	0.9960	0.3240	0.0587	1.3164	0.2611	5.2127	2.2577	0.9643	0.1255	0.0740	1.0815	0.0420	1.6215	2.0151
AETNET	0.9577	0.2669	0.1713	1.1663	0.2450	3.6011	2.1110	0.9716	0.2214	0.0294	1.0513	0.1377	4.5997	2.1341
TDD	0.2809	0.3745	0.2904	0.5736	0.0815	0.7869	1.2023	0.2366	0.0249	0.0611	0.2534	0.1902	0.2966	1.1960
CBAR	0.9939	0.2964	0.0368	1.2895	0.2569	8.0534	2.2483	0.9778	0.1775	0.0342	1.1244	0.1433	5.1901	2.1215
NL2NET	0.9828	0.2054	0.0443	1.3126	0.1640	9.7425	2.1069	0.9745	0.1292	0.0602	1.0110	0.0649	4.6016	2.1568
SSHAD	0.9946	0.3098	0.0341	1.3107	0.2757	9.0850	2.2746	0.9878	0.2076	0.0331	1.1924	0.1745	6.2719	2.1610
SuperAD	0.9261	0.2416	0.0609	1.1578	0.2064	3.9672	2.0541	0.9084	0.0867	0.0835	0.9926	0.0268	1.0383	1.8824
UC-HAD	**0.9971**	**0.3345**	**0.0302**	**1.3343**	**0.3039**	**11.0661**	**2.3039**	**0.9929**	**0.2243**	**0.0290**	**1.2172**	**0.1953**	**7.7345**	**2.1882**

TABLE IV
QUANTITATIVE COMPARISON RESULTS ON THE AVIRIS-I AND HYDICE DATASETS

Model	AVIRIS-I							HYDICE						
	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{JAD}	AUC_{ADBS}	AUC_{SNPR}	AUC_{OADP}	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{JAD}	AUC_{ADBS}	AUC_{SNPR}	AUC_{OADP}
LRX	0.7785	0.1224	0.0439	0.8925	0.0793	2.7919	1.8535	0.9857	0.4419	0.0669	1.4219	0.3797	6.6617	2.3518
CRD	0.9825	0.1651	0.0440	1.1451	0.1294	3.6343	1.7558	0.9942	0.3740	0.0438	1.3639	0.3291	8.6034	2.3246
Auto-AD	0.8726	0.0640	0.0492	0.9342	0.0189	1.2239	1.8850	0.9872	0.3539	0.0527	1.3335	0.2994	6.7338	2.2835
AETNET	0.9798	0.1818	0.0482	1.1617	0.1329	6.3170	2.1329	0.9862	0.3517	0.0753	1.3580	0.2656	6.6722	2.2150
TDD	0.9520	0.0607	0.0490	1.0131	0.0501	6.9576	2.0852	0.8447	0.0837	0.0460	0.9257	0.0308	3.6011	2.0854
CBAR	0.9909	0.1847	0.0424	1.1740	0.1452	4.3561	2.1361	0.9617	0.3024	0.0715	1.2510	0.2368	4.2294	2.1850
NL2NET	0.9819	0.0722	0.0437	1.1242	0.0420	**6.9604**	2.1041	0.9932	0.2620	0.0724	1.3393	0.2416	5.1326	2.1838
SSHAD	0.9887	0.1835	0.0434	1.1732	0.1401	4.2281	2.1325	0.9937	0.3662	0.0572	1.3599	0.3090	6.4021	2.3060
SuperAD	0.9742	0.1586	0.0446	1.1284	0.1139	3.5561	2.0881	0.9920	0.3378	0.0607	1.3241	0.2771	5.5651	2.2692
UC-HAD	**0.9914**	**0.1960**	**0.0415**	**1.1874**	**0.1545**	4.4729	**2.1459**	**0.9951**	**0.3829**	**0.0548**	**1.3781**	**0.3232**	**6.9787**	**2.3232**

results are reported in Tables II–IV. For the primary metric $AUC_{(D,F)}$, UC-HAD achieves the best performance on all six datasets, with an average value of 0.9947. Compared with the second-best method on each dataset, it achieves an average relative improvement of 0.15%, demonstrating its strong anomaly–background discrimination ability. For the threshold-aware metrics, UC-HAD also shows strong and stable performance. Specifically, it obtains average values of 0.2663, 0.0327, and 1.4095 in terms of $AUC_{(D,\tau)}$, $AUC_{(F,\tau)}$, and AUC_{JAD}, respectively. Notably, UC-HAD achieves the lowest $AUC_{(F,\tau)}$ on five of the six datasets, indicating that it can effectively suppress false alarms over varying thresholds while maintaining competitive detection responses. In addition, for AUC_{ADBS}, AUC_{SNPR}, and AUC_{OADP}, UC-HAD obtains average values of 0.2331, 13.2617, and 2.3764, respectively. These results further verify that UC-HAD achieves a favorable balance between enhancing anomalous responses and suppressing background interference. Overall, the proposed method demonstrates superior and consistent performance across different datasets and evaluation criteria.

2) Visualization: In Fig. 5, we present the anomaly detection results of different methods across six public datasets. It can be observed that UC-HAD achieves more accurate localization of anomalous regions compared with competing methods. In particular, the proposed method is able to better preserve the shape and boundary structures of anomalous targets. Meanwhile, it produces fewer false-positive responses in background areas, indicating improved robustness against

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:15:16 UTC from IEEE Xplore. Restrictions apply.

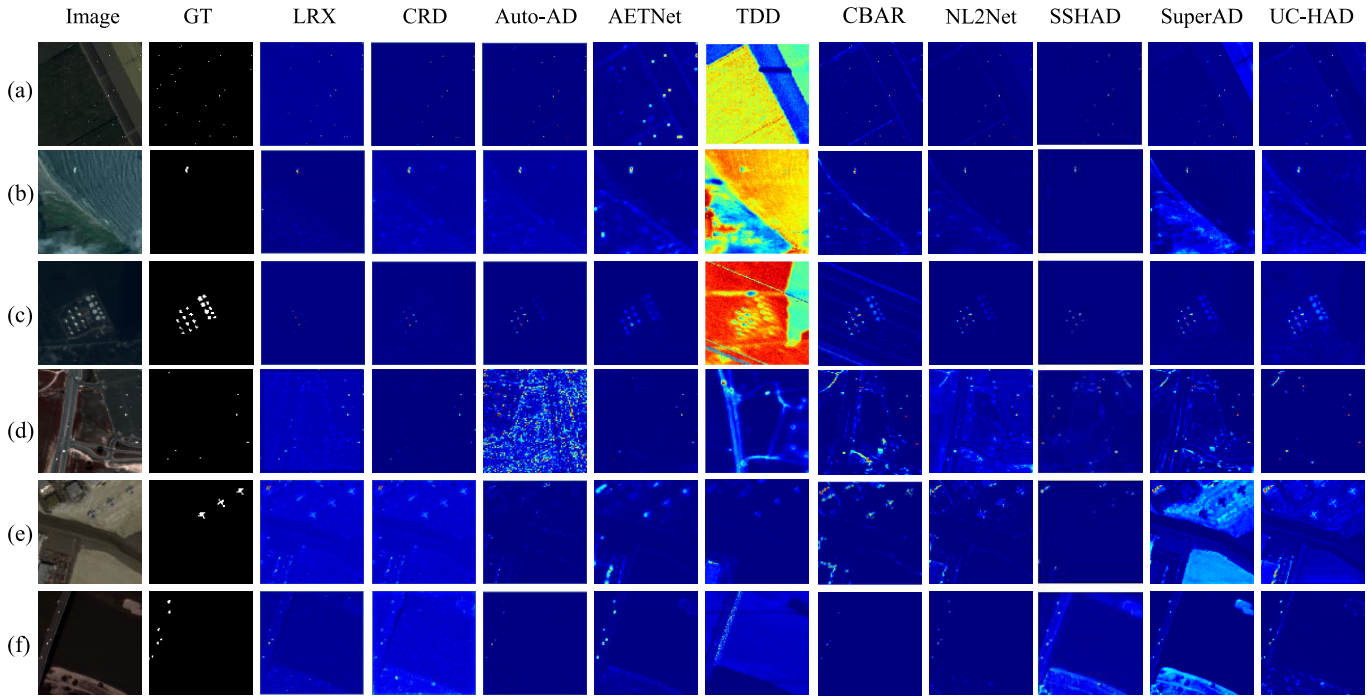


Fig. 5. Detection maps of different methods on the six public HAD datasets. (a) Salinas. (b) Bay Champagne. (c) Texas Coast. (d) HYDICE. (e) HSI-II. (f) AVIRIS-I.

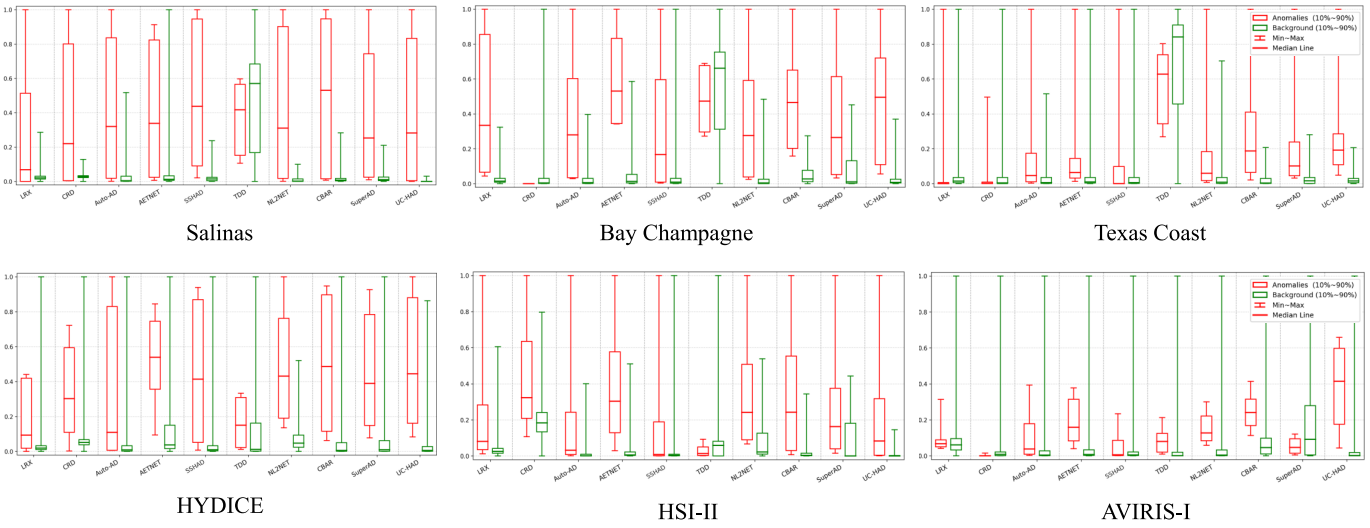


Fig. 6. Separability maps of different methods on the six public HAD datasets.

background interference. Overall, these visual results demonstrate the effectiveness of UC-HAD in anomaly detection tasks.

In Fig. 6, we present the separability maps of different methods across six datasets to visually illustrate the discrimination between anomalous and background regions. Compared with competing methods, UC-HAD exhibits a clearer separation between anomaly and background responses, with anomaly scores (red boxes) shifted toward higher values and background scores (green boxes) concentrated at lower values. The overlap between the two distributions is substantially reduced, indicating stronger anomaly–background discriminability.

In Fig. 7, we visualize the pseudo-background generated by applying the UAFS to the input hyperspectral image. It can be observed that the anomalous regions in the input HSI, which correspond to high-uncertainty areas in the uncertainty map, are effectively repaired into homogeneous components consistent with the surrounding background. Meanwhile, the overall structural characteristics of the scene are well preserved. This result demonstrates that the UAFS can effectively suppress the interference of anomalous components, thereby providing a solid foundation for subsequent pure background modeling.

In Fig. 8, we visualize the ROC curves of different methods across six datasets to evaluate the tradeoff between detection

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:15:16 UTC from IEEE Xplore. Restrictions apply.

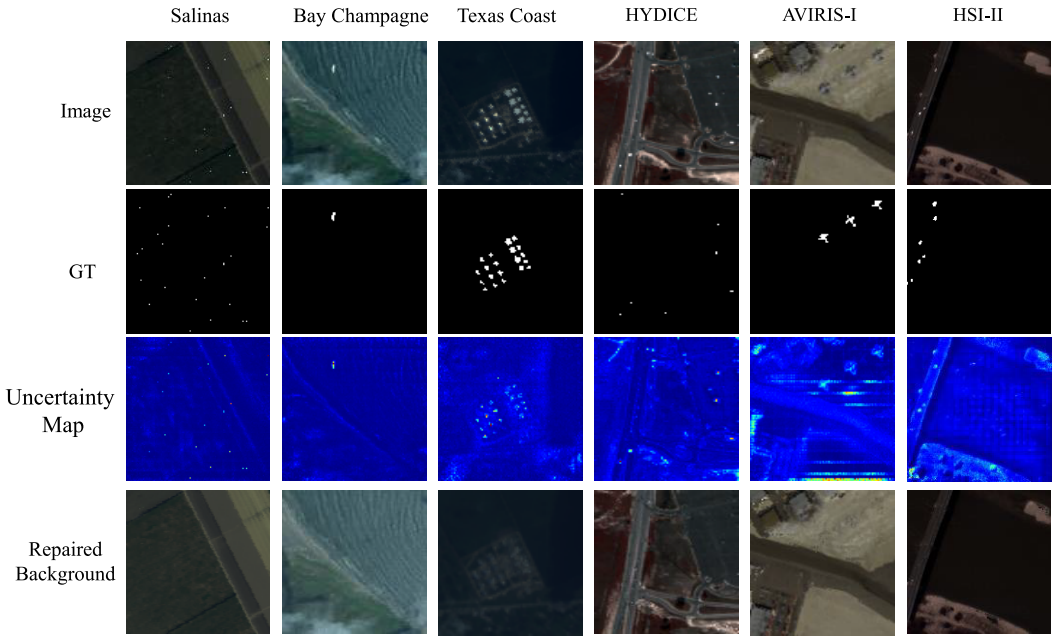


Fig. 7. Visualization of the repaired background generated by the proposed UAFS on the input hyperspectral image.

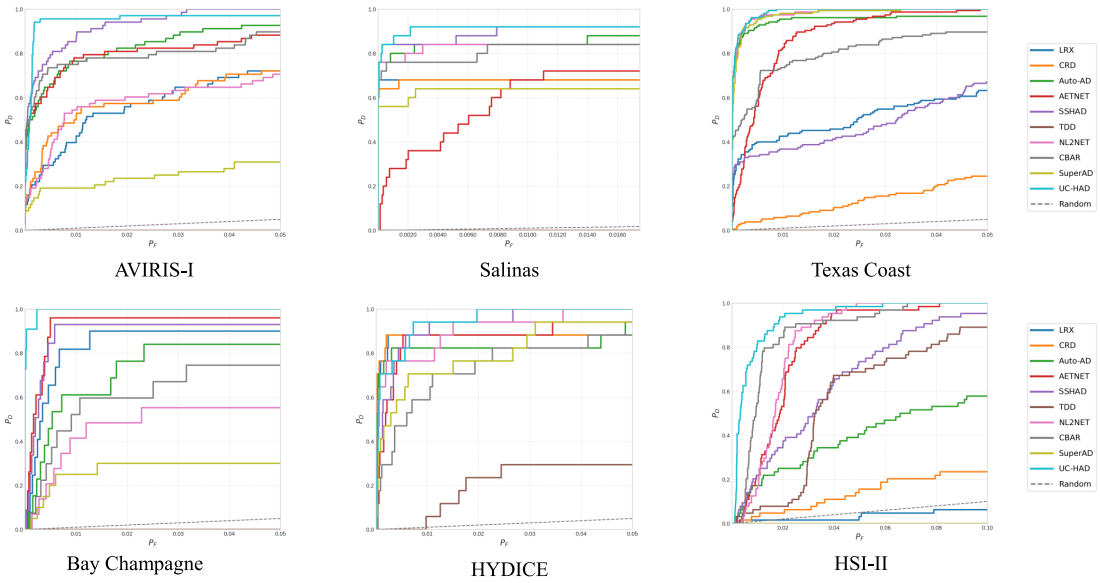


Fig. 8. ROC curves of (D, F) for different methods on the six public HAD datasets.

probability and false-alarm rate. Across most datasets, UC-HAD produces steeper ROC curves and maintains higher detection probabilities under low false-alarm rates, indicating that anomalies can be effectively detected while background false alarms are well suppressed. These results are consistent with the separability maps, demonstrating that the enhanced anomaly–background response separation achieved by UC-HAD can be effectively translated into more reliable detection performance.

In Fig. 9, we visualize two representative detection maps on the additional airborne/satellite hyperspectral datasets, namely UHAD-I and UHAD-II, which contain more complex heterogeneous scenes than the conventional benchmark datasets. UC-HAD accurately highlights anomalous targets while effectively suppressing background responses, demonstrating its robustness in challenging real-world environments. Quantitatively, UC-HAD achieves 98.94% and 99.72% in terms of $\mathrm{AUC}_{(D,F)}$ on UHAD-I and UHAD-II, respectively. More comprehensive qualitative and quantitative results are provided in the Supplementary Material.

In Fig. 10, we visualize the uncertainty maps at different cyclic stages when UC-HAD is applied to the AVIRIS-I dataset. As the number of cycles increases, the proportion of high-uncertainty regions gradually decreases, indicating that unreliable regions are progressively refined during the cyclic process. When the iteration reaches the fifth cycle, the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:15:16 UTC from IEEE Xplore. Restrictions apply.

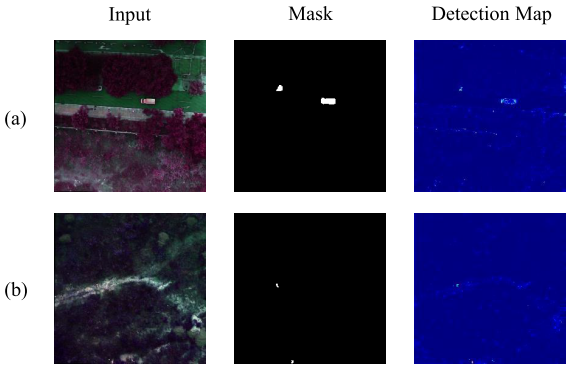


Fig. 9. Detection maps on two additional airborne/satellite hyperspectral datasets with complex heterogeneous scenes. (a) UHAD-I. (b) UHAD-II. More comprehensive qualitative and quantitative results are provided in the Supplementary Material.

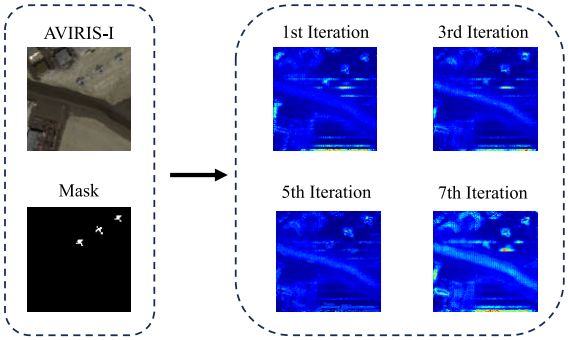


Fig. 10. Visualization of normalized uncertainty maps at different cyclic stages of UC-HAD on the AVIRIS-I dataset. As the number of cycles increases, high-uncertainty responses are progressively suppressed, while excessive iterations may lead to over-refinement. Note that each uncertainty map is independently normalized within its corresponding cycle. Therefore, the highlighted regions indicate relatively high reconstruction instability in the current cycle rather than absolute reconstruction failure or exact anomaly locations.

high-uncertainty regions become very small and are mainly concentrated around anomaly-related regions, suggesting that anomalous interference has been effectively suppressed and the reconstructed result can be regarded as an approximate background representation. However, when the number of cycles is further increased to 7, the uncertainty map exhibits a re-emergence of spatially diffused high-uncertainty responses. This phenomenon indicates that the cyclic process has exceeded the effective background purification stage. After repeated UAFS repair and BR reconstruction, the input fed into the UE has been largely transformed into background-like components, and the contrast between anomalous and background regions is gradually weakened. As a result, the uncertainty responses are no longer concentrated on anomaly-related unreliable regions, but become more dispersed over background areas. This suggests that excessive cyclic iterations may reduce the discriminative guidance of uncertainty and lead to over-refinement, thereby weakening anomaly–background separability.

C. Model Analysis

In this section, we systematically conduct ablation studies to analyze the effects of different reconstruction reliability modeling mechanisms, filling strategies, and detection paradigms on model performance. In addition, we further investigate the convergence behavior and model parameter scale of UC-HAD on the overall performance.

TABLE V
COMPARISON OF DIFFERENT RECONSTRUCTION RELIABILITY MODELING MECHANISMS IN $AUC_{(D,F)}$ METRIC

Methods	Salinas	Texas Coast	Bay Champagne	HSI-II	AVIRIS-I	HYDICE
Handcrafted prior	0.7335	0.6824	0.8369	0.7812	0.6602	0.8730
Bayesian perception	0.9059	**0.9991**	0.8915	0.8047	0.8817	0.8808
MC-Dropout	**0.9926**	0.9988	**0.9971**	**0.9929**	**0.9914**	**0.9951**

TABLE VI
COMPARISON OF DIFFERENT FILLING STRATEGIES IN THE $AUC_{(D,F)}$ METRIC

Methods	Salinas	Texas Coast	Bay Champagne	HSI-II	AVIRIS-I	HYDICE
No Filling	0.9033	0.9804	0.9833	0.7734	0.8566	0.9343
Mean Filling	0.9177	0.9896	0.9407	0.8042	0.8998	0.9740
UAFS	**0.9926**	**0.9988**	**0.9971**	**0.9929**	**0.9914**	**0.9951**

TABLE VII
COMPARISON OF DIFFERENT DETECTION FRAMEWORKS IN THE $AUC_{(D,F)}$ METRIC

Methods	Salinas	Texas Coast	Bay Champagne	HSI-II	AVIRIS-I	HYDICE
UE only	0.8977	0.9028	0.7572	0.9655	0.8963	0.8001
BR only	0.8742	0.8184	0.8742	0.9651	0.8721	0.8643
One-pass	0.9468	0.9635	0.9304	0.9748	0.9496	0.9347
Cyclic	**0.9926**	**0.9988**	**0.9971**	**0.9929**	**0.9914**	**0.9951**

1) Reliability Modeling Mechanisms: We compare the effects of three different reliability modeling mechanisms on the detection performance of UC-HAD, as reported in Table V. Here, a reconstruction reliability modeling mechanism refers to the method used to generate uncertainty maps within the UC-HAD framework.

First, under the handcrafted prior method, we adopt the classical RX-style framework. Specifically, for a given pixel $\mathbf{x}$, its local statistics are estimated from the surrounding spatial context. The uncertainty is quantified as the Mahalanobis distance between the pixel and the mean of the corresponding local distribution. This method achieves only an average $AUC_{(D,F)}$ of 76.12% across different datasets. The relatively limited performance suggests that handcrafted statistical modeling alone is insufficient to provide robust reconstruction reliability estimation in complex hyperspectral scenes.

Second, under the Bayesian reliability modeling strategy, each convolutional layer in the UE is implemented as a Bayesian convolutional layer, where the posterior distribution of the weights is approximated by a Gaussian distribution. During forward propagation, the reparameterization trick is employed to sample the weights, enabling weight uncertainty to propagate to the output and thereby quantifying predictive uncertainty at each spatial location. This strategy achieves an average $AUC_{(D,F)}$ of 89.40% over six datasets. However, its main drawback lies in the substantial computational overhead,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:15:16 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII
PARAMETER SENSITIVITY ANALYSIS OF KEY HYPERPARAMETERS IN UAFS IN TERMS OF $AUC_{(D,F)}$. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Parameter	Value	Salinas	Texas Coast	Bay Champagne	HSI-II	AVIRIS-I	HYDICE	Avg.
σ_{spec}	0.05	0.9785	0.9912	0.9836	0.9633	0.9725	0.9867	0.9793
	0.10	0.9889	0.9967	0.9938	0.9851	0.9876	0.9928	0.9908
	0.15	**0.9926**	**0.9988**	**0.9971**	**0.9929**	**0.9914**	**0.9951**	**0.9947**
	0.20	0.9894	0.9961	0.9925	0.9827	0.9859	0.9922	0.9898
	0.25	0.9796	0.9904	0.9821	0.9615	0.9705	0.9860	0.9784
α	0.2	0.9768	0.9916	0.9827	0.9634	0.9719	0.9861	0.9788
	0.4	0.9884	0.9972	0.9940	0.9863	0.9869	0.9924	0.9909
	0.6	**0.9926**	**0.9988**	**0.9971**	**0.9929**	**0.9914**	**0.9951**	**0.9947**
	0.8	0.9691	0.9968	0.9935	0.9848	0.9857	0.9920	0.9870
	1.0	0.9712	0.9898	0.9795	0.9568	0.9682	0.9843	0.9750
Top-K	10	0.9845	0.9948	0.9906	0.9817	0.9829	0.9904	0.9875
	15	0.9897	0.9972	0.9945	0.9876	0.9878	0.9931	0.9917
	20	**0.9926**	**0.9988**	**0.9971**	**0.9929**	**0.9914**	**0.9951**	**0.9947**
	25	0.9901	0.9975	0.9949	0.9884	0.9881	0.9935	0.9921
	30	0.9832	0.9939	0.9893	0.9768	0.9897	0.9887	0.9869

particularly due to repeated sampling during inference, which significantly increases the overall inference time.

Finally, the MC-Dropout mechanism attains an average $\text{AUC}_{(D,F)}$ of 99.47% across the six datasets, outperforming the aforementioned strategies while maintaining a relatively moderate computational cost, thus achieving a more balanced tradeoff between detection accuracy and efficiency.

2) Filling Strategy: We evaluate the impact of three different filling strategies on the detection performance of UC-HAD, as shown in Table VI. When no filling strategy is applied, that is, the original HSI is directly fed into the reconstructor, UC-HAD achieves an average $\text{AUC}_{(D,F)}$ of 90.52% across the datasets. When the mean filling strategy is adopted, that is, the corresponding regions are filled using the spectral mean of local pixels surrounding high-uncertainty areas, the average $\text{AUC}_{(D,F)}$ of UC-HAD increases to 92.10%. Furthermore, when the UAFS is employed, UC-HAD attains an average $\text{AUC}_{(D,F)}$ of 99.47%, which is significantly superior to the previous two strategies. These results demonstrate that the UAFS effectively suppresses the interference of anomalous components in the background modeling process, thereby improving the overall detection performance.

3) Detection Framework: To validate the effectiveness of the proposed cyclic framework and further analyze the interaction between the UE and BR, we compare different UE–BR interaction mechanisms, as reported in Table VII. Specifically, four settings are evaluated, including UE-only, BR-only, one-pass UE–UAFS–BR, and cyclic UE–UAFS–BR. In the UE-only setting, the uncertainty map generated by the UE is directly taken as the final detection output, yielding an average $\text{AUC}_{(D,F)}$ of 86.99% across the datasets. In the BR-only setting, the reconstruction residual map produced by the BR is adopted as the anomaly representation for evaluation, achieving an average $\text{AUC}_{(D,F)}$ of 87.81%. For the one-pass UE–UAFS–BR setting, the UE first estimates the uncertainty map, UAFS repairs the high-uncertainty regions once, and the BR then reconstructs the background without further cyclic updates. This setting achieves an average $\text{AUC}_{(D,F)}$ of 94.99%, indicating that uncertainty-guided background repair is effective for improving reconstruction-based anomaly detection. In contrast, the complete cyclic framework achieves the best detection performance on all six public HAD datasets, with an average $\text{AUC}_{(D,F)}$ of 99.47%. These results demonstrate that the performance gain of UC-HAD does not come from a simple combination of the UE and BR, but from their cyclic interaction. Through this mutual reinforcement, the cyclic framework progressively improves background modeling quality and significantly enhances anomaly detection performance.

4) Hyperparameter Analysis: To evaluate the robustness of UC-HAD with respect to the key hyperparameters in UAFS, we conduct *one-factor-at-a-time* sensitivity experiments on six public HAD datasets. Specifically, when analyzing one hyperparameter, the remaining hyperparameters are fixed to their default values. Table VIII presents the parameter sensitivity results in terms of $\text{AUC}_{(D,F)}$. It can be observed that UC-HAD maintains stable detection performance within a reasonable parameter range. For the spectral similarity scale σ_{spec}, a relatively small value imposes a strict spectral similarity constraint and may reduce the number of effective background candidates, whereas a large value may introduce spectrally dissimilar pixels into the filling process. The setting $\sigma_{\text{spec}} = 0.15$ provides a favorable balance across different datasets. For the fusion factor α, the best overall performance is generally achieved around $\alpha = 0.6$, suggesting that combining global reliable background evidence with local spatial context is more effective than relying on either component alone. For the number of selected candidates K, insufficient candidates may lead to unstable background estimation, whereas too many candidates may introduce less reliable samples. The setting $K = 20$ achieves consistently competitive results. In addition, the sensitivity analyses of the local neighborhood radius r and the number of stochastic forward passes T are provided in the Supplementary Material.

V. CONCLUSION AND DISCUSSION

In this article, we revisited reconstruction-based HAD from the perspective of background reconstruction reliability and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:15:16 UTC from IEEE Xplore. Restrictions apply.

addressed the reliance of masking-based methods on predefined occlusion patterns. Building upon this perspective, we proposed an uncertainty-driven cyclic framework (UC-HAD), where pixel-wise reconstruction uncertainty is introduced as a data-adaptive cue to guide background purification. Within UC-HAD, uncertainty estimation and background reconstruction interact cyclically, enabling iterative refinement of the reconstructed background. Extensive experiments on six datasets demonstrate that UC-HAD consistently achieves strong detection performance, validating the effectiveness of the proposed uncertainty-guided cyclic background purification framework. Nevertheless, some limitations remain. A small number of heterogeneous background structures may be mistakenly identified as high-uncertainty regions, and excessive purification may weaken anomaly–background contrast, thereby degrading detection performance. Therefore, our future work will explore more structure-aware repair strategies to enhance the robustness of UC-HAD across diverse and heterogeneous scenes.

REFERENCES

[1] H. Su, Z. Wu, H. Zhang, and Q. Du, "Hyperspectral anomaly detection: A survey," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 1, pp. 64–90, Mar. 2022.

[2] Y. Xu, L. Zhang, B. Du, and L. Zhang, "Hyperspectral anomaly detection based on machine learning: An overview," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 3351–3364, 2022.

[3] J. Li, X. Wang, S. Wang, H. Zhao, and Y. Zhong, "One-step detection paradigm for hyperspectral anomaly detection via spectral deviation relationship learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5517515.

[4] I. S. Reed and X. Yu, "Adaptive multiple-band CFAR detection of an optical pattern with unknown spectral distribution," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 38, no. 10, pp. 1760–1770, Oct. 1990.

[5] S. Matteoli, M. Diani, and G. Corsini, "A tutorial overview of anomaly detection in hyperspectral images," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 25, no. 7, pp. 5–28, Jul. 2010.

[6] J. Lei, W. Xie, J. Yang, Y. Li, and C.-I. Chang, "Spectral–spatial feature extraction for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 8131–8143, Oct. 2019.

[7] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[8] H. Petersson, D. Gustafsson, and D. Bergstrom, "Hyperspectral image analysis using deep learning—A review," in *Proc. 6th Int. Conf. Image Process. Theory, Tools Appl. (IPTA)*, Dec. 2016, pp. 1–6.

[9] W. Chen, X. Zhi, S. Jiang, Y. Huang, Q. Han, and W. Zhang, "DWSDiff: Dual-window spectral diffusion for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5504617.

[10] J. Lian, L. Wang, H. Sun, and H. Huang, "GT-HAD: Gated transformer for hyperspectral anomaly detection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 2, pp. 3631–3645, Feb. 2025.

[11] S. Wang, X. Wang, L. Zhang, and Y. Zhong, "Auto-AD: Autonomous hyperspectral anomaly detection network based on fully convolutional autoencoder," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

[12] J. Lei, S. Fang, W. Xie, Y. Li, and C.-I. Chang, "Discriminative reconstruction for hyperspectral anomaly detection with spectral learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7406–7417, Oct. 2020.

[13] T. Jiang, Y. Li, W. Xie, and Q. Du, "Discriminative reconstruction constrained generative adversarial network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4666–4679, Jul. 2020.

[14] Z. Li, Y. Wang, C. Xiao, Q. Ling, Z. Lin, and W. An, "You only train once: Learning a general anomaly enhancement network with random masks for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5506718.

[15] Q. Guo, Y. Cen, L. Zhang, Y. Zhang, and Y. Huang, "Hyperspectral anomaly detection based on spatial–spectral cross-guided mask autoencoder," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 9876–9889, 2024.

[16] Z. Yang et al., "A multi-scale mask convolution-based blind-spot network for hyperspectral anomaly detection," *Remote Sens.*, vol. 16, no. 16, p. 3036, Aug. 2024.

[17] D. Wang, L. Zhuang, L. Gao, X. Sun, M. Huang, and A. Plaza, "BockNet: Blind-block reconstruction network with a guard window for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5531916.

[18] Z. Wu et al., "SMCNet: Sparse-inspired masked convolutional network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5535317.

[19] Z. Jiang et al., "Cross-domain hyperspectral image classification," *Pattern Recognit.*, vol. 168, Feb. 2025, Art. no. 111836.

[20] S. Feng et al., "Transformer-based cross-domain few-shot learning for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5501716.

[21] F. Li et al., "Structured background modeling for hyperspectral anomaly detection," *Sensors*, vol. 18, no. 9, p. 3137, Sep. 2018.

[22] J. Liu, Z. Hou, W. Li, R. Tao, D. Orlando, and H. Li, "Multipixel anomaly detection with unknown patterns for hyperspectral imagery," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 10, pp. 5557–5567, Oct. 2022.

[23] L. Zhuang and M. K. Ng, "FastHyMix: Fast and parameter-free hyperspectral image mixed noise removal," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 8, pp. 4702–4716, Aug. 2023.

[24] L. Zhuang, M. K. Ng, X. Fu, and J. M. Bioucas-Dias, "Hy-demosaicing: Hyperspectral blind reconstruction from spectral subsampling," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5515815.

[25] C. Zhao, C. Li, S. Feng, N. Su, and W. Li, "A spectral–spatial anomaly target detection method based on fractional Fourier transform and saliency weighted collaborative representation for hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 5982–5997, 2020.

[26] L. Zhuang and M. K. Ng, "Hyperspectral mixed noise removal by ℓ_1-norm-based subspace representation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 1143–1157, 2020.

[27] C.-I. Chang, "Target-to-anomaly conversion for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5540428.

[28] P. Gurram and H. Kwon, "Support-vector-based hyperspectral anomaly detection using optimized kernel parameters," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 6, pp. 1060–1064, Nov. 2011.

[29] W.-M. Liu and C.-I. Chang, "Multiple-window anomaly detection for hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 2, pp. 644–658, Apr. 2013.

[30] W. Xie, T. Jiang, Y. Li, X. Jia, and J. Lei, "Structure tensor and guided filtering-based algorithm for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4218–4230, Jul. 2019.

[31] C.-I. Chang, "Hyperspectral target detection: Hypothesis testing, signal-to-noise ratio, and spectral angle theories," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5505223.

[32] H. Kwon and N. M. Nasrabadi, "Kernel RX-algorithm: A nonlinear anomaly detector for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 2, pp. 388–397, Feb. 2005.

[33] Q. Guo, B. Zhang, Q. Ran, L. Gao, J. Li, and A. Plaza, "Weighted-RXD and linear filter-based RXD: Improving background statistics estimation for anomaly detection in hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2351–2366, Jun. 2014.

[34] L. Zhuang, M. K. Ng, and Y. Liu, "Cross-track illumination correction for hyperspectral pushbroom sensor images using low-rank and sparse representations," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5502117.

[35] L. Ren, Z. Ma, and F. Bovolo, "A novel dual-alternating direction method of multipliers for spectral unmixing," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 3, pp. 528–532, Mar. 2021.

[36] F. Li, Y. Zhang, L. Zhang, X. Zhang, and D. Jiang, "Hyperspectral anomaly detection using background learning and structured sparse representation," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 1618–1621.

[37] Z. Wu et al., "Hyperspectral anomaly detection with relaxed collaborative representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5533417.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:15:16 UTC from IEEE Xplore. Restrictions apply.

[38] Y. Zhang, B. Du, L. Zhang, and S. Wang, "A low-rank and sparse matrix decomposition-based Mahalanobis distance method for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1376–1389, Mar. 2016.

[39] J. Wang, S. Guo, Z. Hua, R. Huang, J. Hu, and M. Gong, "CL-CaGAN: Capsule differential adversarial continual learning for cross-domain hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 6001174.

[40] C. Huang, L. Liu, C. Xiao, Q. Ling, G. Li, and K. Li, "Endmember-guided spectral–spatial feature diffusion for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, doi: 10.1109/TGRS.2025.3641223.

[41] Z. Wu and B. Wang, "Transformer-based autoencoder framework for nonlinear hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5508015.

[42] S. Xiao, T. Zhang, Z. Xu, J. Qu, S. Hou, and W. Dong, "Anomaly detection of hyperspectral images based on transformer with spatial–spectral dual-window mask," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 1414–1426, 2023.

[43] T. Guo, Y. Yang, F. Luo, C. Fu, and L. Zhang, "D3Former: Dual-decoder dual-transformer reconstruction network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5528215.

[44] X. Fu, T. Zhang, J. Cheng, and S. Jia, "MMR-HAD: Multiscale mamba reconstruction network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5516914.

[45] X. He et al., "CWIMamba: Cross-scale windowed integration state space model for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5527820.

[46] D. Hong et al., "SpectralGPT: Spectral remote sensing foundation model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5227–5244, Aug. 2024.

[47] W. Kong, B. Liu, X. Bi, C. Yu, X. Li, and Y. Chen, "HyperSL: A spectral foundation model for hyperspectral image interpretation," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5513119.

[48] J. Li et al., "HyperFree: A channel-adaptive and tuning-free foundation model for hyperspectral remote sensing imagery," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 23048–23058.

[49] D. Wang et al., "HyperSIGMA: Hyperspectral intelligence comprehension foundation model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 8, pp. 6427–6444, Aug. 2025.

[50] F. Luo, S. Shi, K. Qin, T. Guo, C. Fu, and Z. Lin, "SelfMTL: Self-supervised meta-transfer learning via contrastive representation for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5508613.

[51] Q. Wu, J. Li, N. Wang, and J. Peng, "TuckerNet: A multilayer unsupervised hyperspectral change detection framework with tensor decomposition," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 19, pp. 11388–11400, 2026.

[52] L. Wu et al., "DIEFEN: Differential information-enhanced feature exchange network for hyperspectral change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5533312.

[53] E. Hüllermeier and W. Waegeman, "Aleatoric and epistemic uncertainty in machine learning: An introduction to concepts and methods," *Mach. Learn.*, vol. 110, no. 3, pp. 457–506, Mar. 2021.

[54] F. Cuzzolin, "Uncertainty measures: A critical survey," *Inf. Fusion*, vol. 114, Feb. 2025, Art. no. 102609.

[55] A. Kendall and Y. Gal, "What uncertainties do we need in Bayesian deep learning for computer vision?" in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 5574–5584.

[56] B. Lakshminarayanan, A. Pritzel, and C. Blundell, "Simple and scalable predictive uncertainty estimation using deep ensembles," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 6402–6413.

[57] Y. Gal and Z. Ghahramani, "Dropout as a Bayesian approximation: Representing model uncertainty in deep learning," in *Proc. Int. Conf. Mach. Learn.*, 2016, pp. 1050–1059.

[58] P. Ebel, V. S. F. Garnot, M. Schmitt, J. D. Wegner, and X. X. Zhu, "UnCRtainTS: Uncertainty quantification for cloud removal in optical satellite time series," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2023, pp. 2086–2096.

[59] C. Huang, Q. Xu, Y. Wang, Y. Wang, and Y. Zhang, "Self-supervised masking for unsupervised anomaly detection and localization," *IEEE Trans. Multimedia*, vol. 25, pp. 4426–4438, 2023.

[60] A. Pant, A. A. Baniya, T.-K. Lee, and S. Aryal, "Hyperspectral anomaly detection methods: A survey and comparative study," 2025, *arXiv:2507.05730*.

[61] J. M. Molero, E. M. Garzón, I. García, E. S. Quintana-Ortí, and A. Plaza, "Efficient implementation of hyperspectral anomaly detection techniques on GPUs and multicore processors," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2256–2266, Jun. 2014.

[62] H. Liu, X. Su, X. Shen, and X. Zhou, "MSNet: Self-supervised multiscale network with enhanced separation training for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5520313.

[63] Y. Zeng, X.-L. Zhao, T.-Y. Ji, W.-H. Wu, D. Wang, and L. Zhuang, "Continuous tensor representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5520515.

[64] D. Wang, L. Ren, X. Sun, L. Gao, and J. Chanussot, "Nonlocal and local feature-coupled self-supervised network for hyperspectral anomaly detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 6981–6993, 2025.

[65] Y. Cui, J. Zhang, Z. Yan, P. Liu, W. Song, and Y. Zeng, "Overcoming the identity mapping problem in self-supervised hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5514913.

[66] S. Liu, L. Peng, X. Chang, Z. Wang, G. Wen, and C. Zhu, "Adaptive dual-domain learning for hyperspectral anomaly detection with state-space models," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5503719.

[67] C.-I. Chang, "Comprehensive analysis of receiver operating characteristic (ROC) curves for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5541124.

[68] C.-I. Chang, S. Chen, S. Zhong, and Y. Shi, "Exploration of data scene characterization and 3D ROC evaluation for hyperspectral anomaly detection," *Remote Sens.*, vol. 16, no. 1, p. 135, Dec. 2023.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:15:16 UTC from IEEE Xplore. Restrictions apply.